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Total Points
11 / 11 pts

Question 1

Q45 / 5 pts

+ 0 pts Incorrect

✓ + 5 pts Correct

Question 2

Q56 / 6 pts

5 (a)

+ 0 pts 5 (a) Completely Incorrect or Unattempted

✓ + 3 pts 5 (a) Completely Correct

+ 1 pt 5 (a) Partially Correct

+ 2 pts 5 (a) Partially Correct

5 (b)

+ 0 pts 5 (b) Completely Incorrect or Unattempted

✓ + 3 pts 5 (b) Completely Correct

+ 1.5 pts 5 (b) Partially Correct

4. Let $(\mathbb{R}, d)$ be the usual metric spaces, and $f, f_n : \mathbb{R} \rightarrow \mathbb{R}$ be such that $\{f_n\}$ converges uniformly to f on $\mathbb{R}$. Show that $D(f) \subset \bigcup_{n=1}^{\infty} D(f_n)$, where $D(f)$ is the set of discontinuous points of f . [5]

~~$\{f_n\}$ converges uniformly.~~

$\Rightarrow \forall \epsilon > 0, \exists N_\epsilon > 0$ s.t.

$|f_n(x) - f(x)| < \epsilon$ whenever $n \geq N_\epsilon, \forall x \in \mathbb{R}$.

take $\epsilon > 0$.

$\exists N_\epsilon$ s.t. $|f_n(x) - f(x)| < \epsilon$
 $\forall n \geq N_\epsilon, \forall x \in \mathbb{R}$.

~~take the pt~~ take x_0 where
 f is continuous.

now

to show $D(f) \subset \bigcup_{n=1}^{\infty} D(f_n)$

is equivalent to showing

$$(D(f))^c \supset \bigcap_{n=1}^{\infty} (D(f_n))^c$$

suppose $x_0 \in \bigcap_{n=1}^{\infty} (D(f_n))^c$

that is x is continuous
for f_n is cont. at x
 $\forall n$.

take $\epsilon > 0$.

$\exists N_\epsilon > 0$ s.t. $|f_n(x) - f(x)| < \frac{\epsilon}{3}$
 $\forall n \geq N_\epsilon, \forall x \in \mathbb{R}$

$$|f_{N_\epsilon}(y) - f_{N_\epsilon}(x_0)| < \frac{\epsilon}{3}$$

whenever $d(y, x_0) < \delta$

now $|f(y) - f(x_0)|$ $\hookrightarrow$ normal metric.

$$\begin{aligned} &= |f(y) - f_{N_\epsilon}(y) + f_{N_\epsilon}(y) - f_{N_\epsilon}(x_0) \\ &\quad + f_{N_\epsilon}(x_0) - f(x_0)| \\ &\leq |f(y) - f_{N_\epsilon}(y)| + |f_{N_\epsilon}(y) - f_{N_\epsilon}(x_0)| \\ &\quad + |f_{N_\epsilon}(x_0) - f(x_0)| \end{aligned}$$

$$< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

$$\Rightarrow |f(y) - f(x_0)| < \epsilon$$

whenever $|y - x_0| < \delta$

hence f is continuous

at x_0

$$\Rightarrow x_0 \in (D(f))^c$$

hence proved.

..... Rough Work

at pt x .

$$|f_n(x_m) - f_n(x)|$$

$$|f_n(x_m) - f_n(x)| < \epsilon$$

when $d(y, x) < \delta_{n, \epsilon}$

$$\begin{aligned} \text{now } |f(y) - f(x)| &< |f_n(y) - f_n(x)| \\ &< |f_n(y) - f_n(x)| \\ &\quad + |f_n(x) - f(x)| \\ &\quad + |f(x) - f(y)| \end{aligned}$$

$$|f(y) - f(x)| < \epsilon$$

if it is continuous at $x \rightarrow$

$$|S_n(x) - S_n(x)|$$

$$S_n = \sum_{i=1}^n g_i \geq \sum_{i=1}^n \frac{1}{n^2}$$

$\hookrightarrow$ converges pointwise.

$$|S_n(x) - S_m(x)| < \epsilon \quad \forall n, m \in \mathbb{N}$$

take $B_{\epsilon}(x)$

also since f_{n_ϵ} is cont at x_0
 $\exists \delta_{\epsilon, n_0}$

5

6

$n \in B_\epsilon(n)$.

$\overline{n_1, \dots, n_m}$.